

Structured light 3D scanning in the presence of global illumination

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Context & Relevance

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- Circuit boards
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 - How can you get 3D structure of circuit board from 2D images?

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Structured light 3D scanners!

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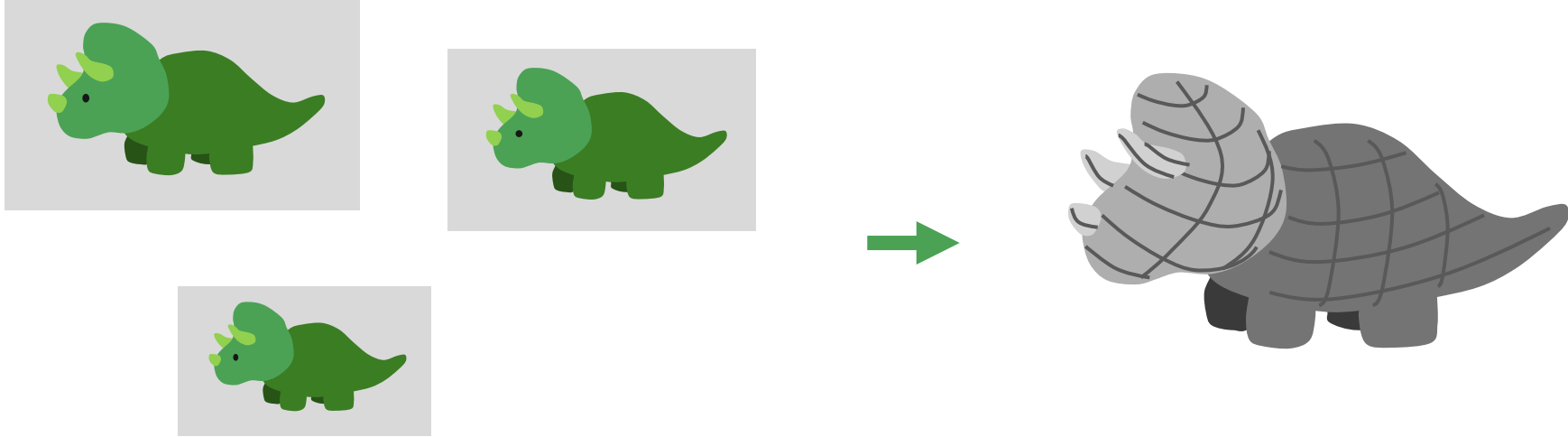
Structured light 3D scanners!

General idea: project light onto a scene, take pictures of the scene, and use image pixels to help compute the depth of each pixel's corresponding scene point

Overall structure of this presentation

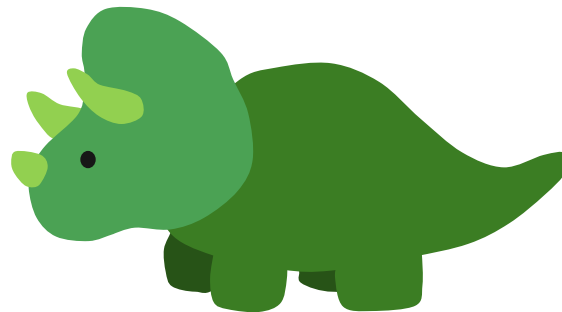
- Overview of **structured light 3D scanning**
- Description of **global illumination** and the **errors** it can cause in structured light 3D scanning
- Explanation of how these errors are **prevented, detected and corrected**

How to recover 3D structure of a scene from 2D images?



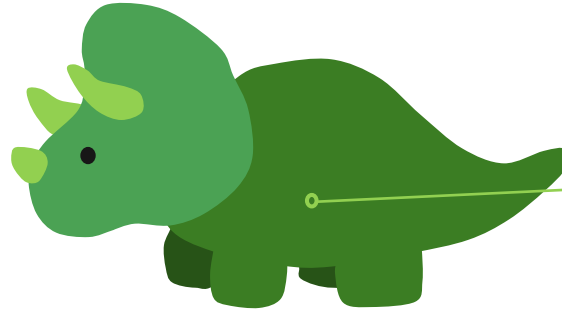
Simple stereo

Object S



Simple stereo

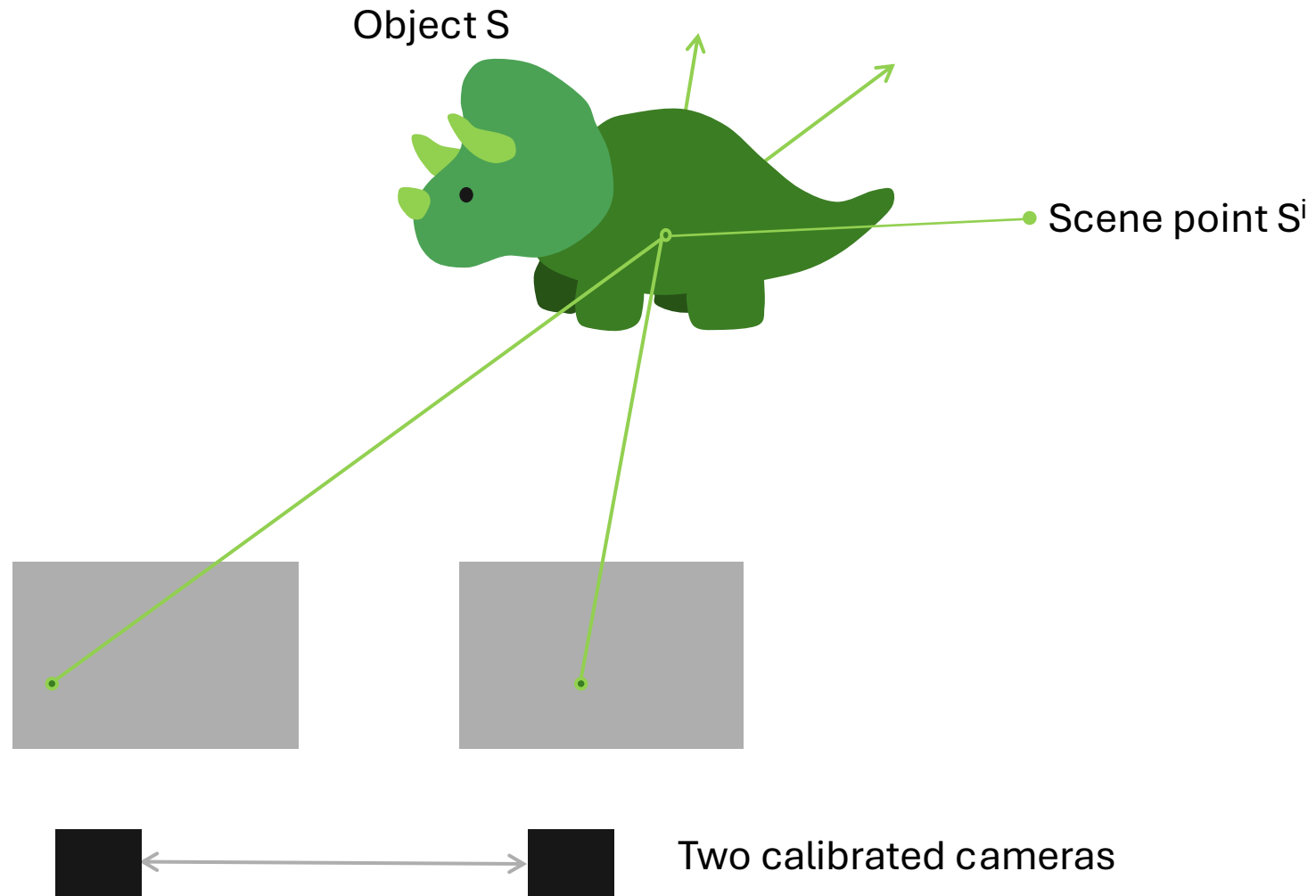
Object S



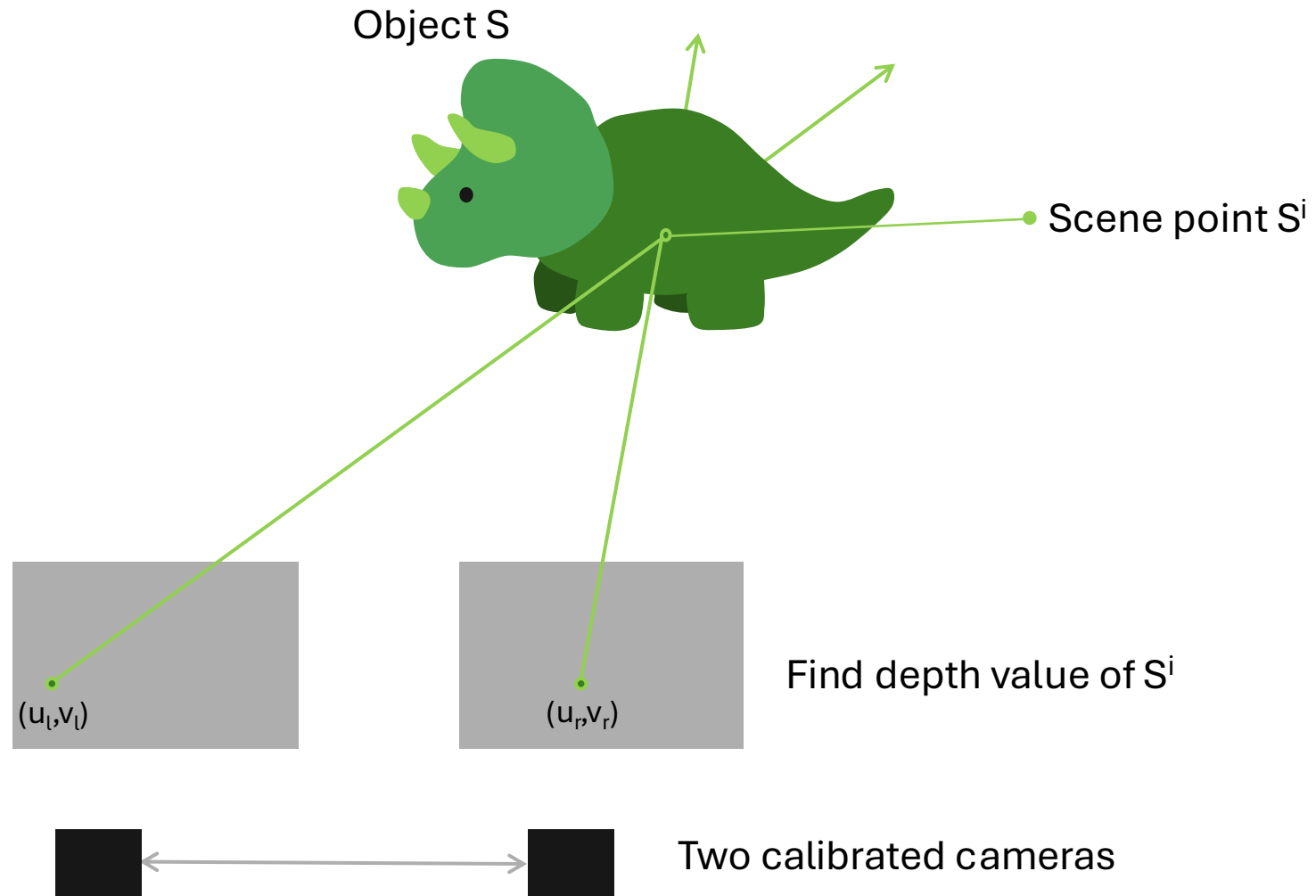
Scene point S^i



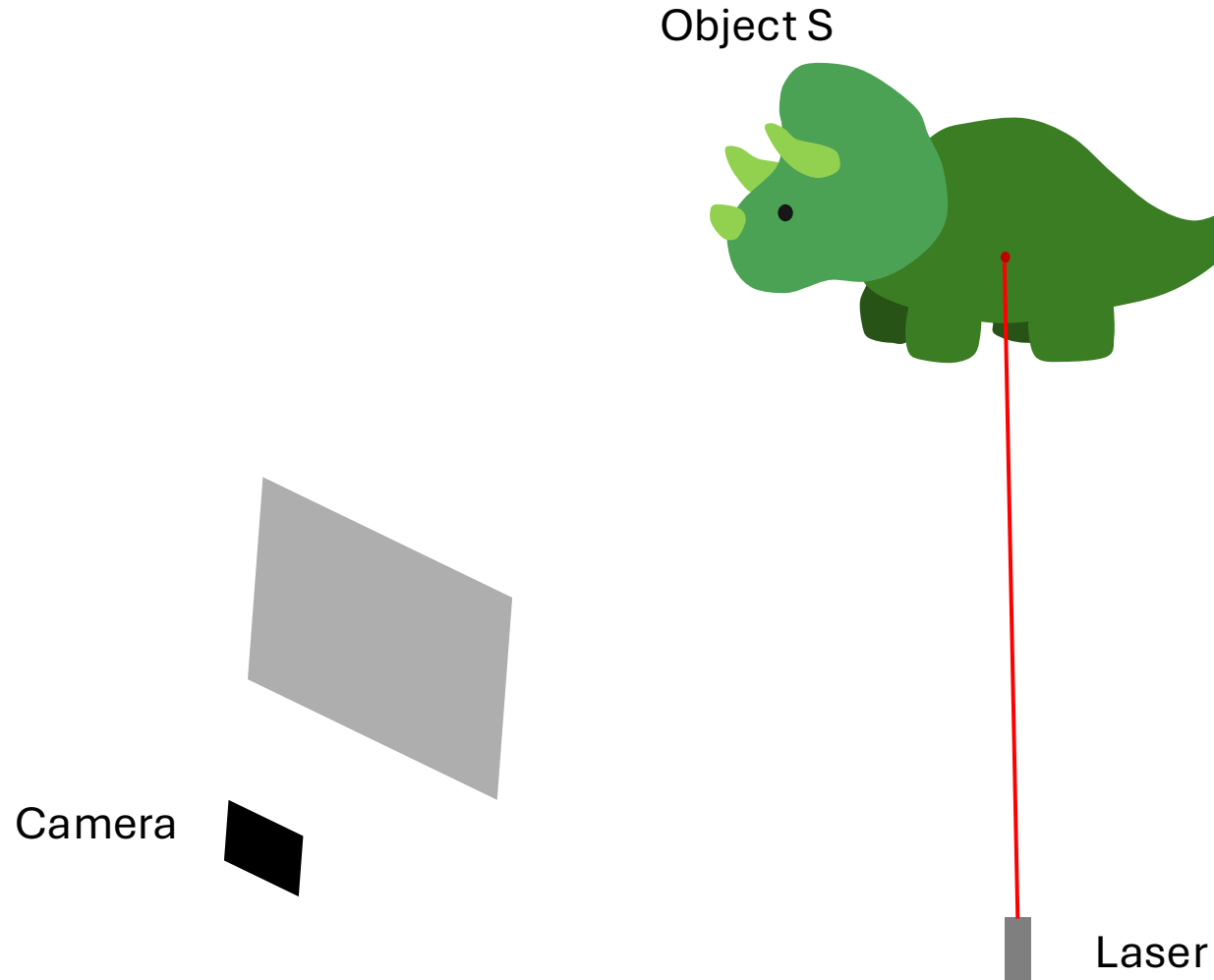
Simple stereo



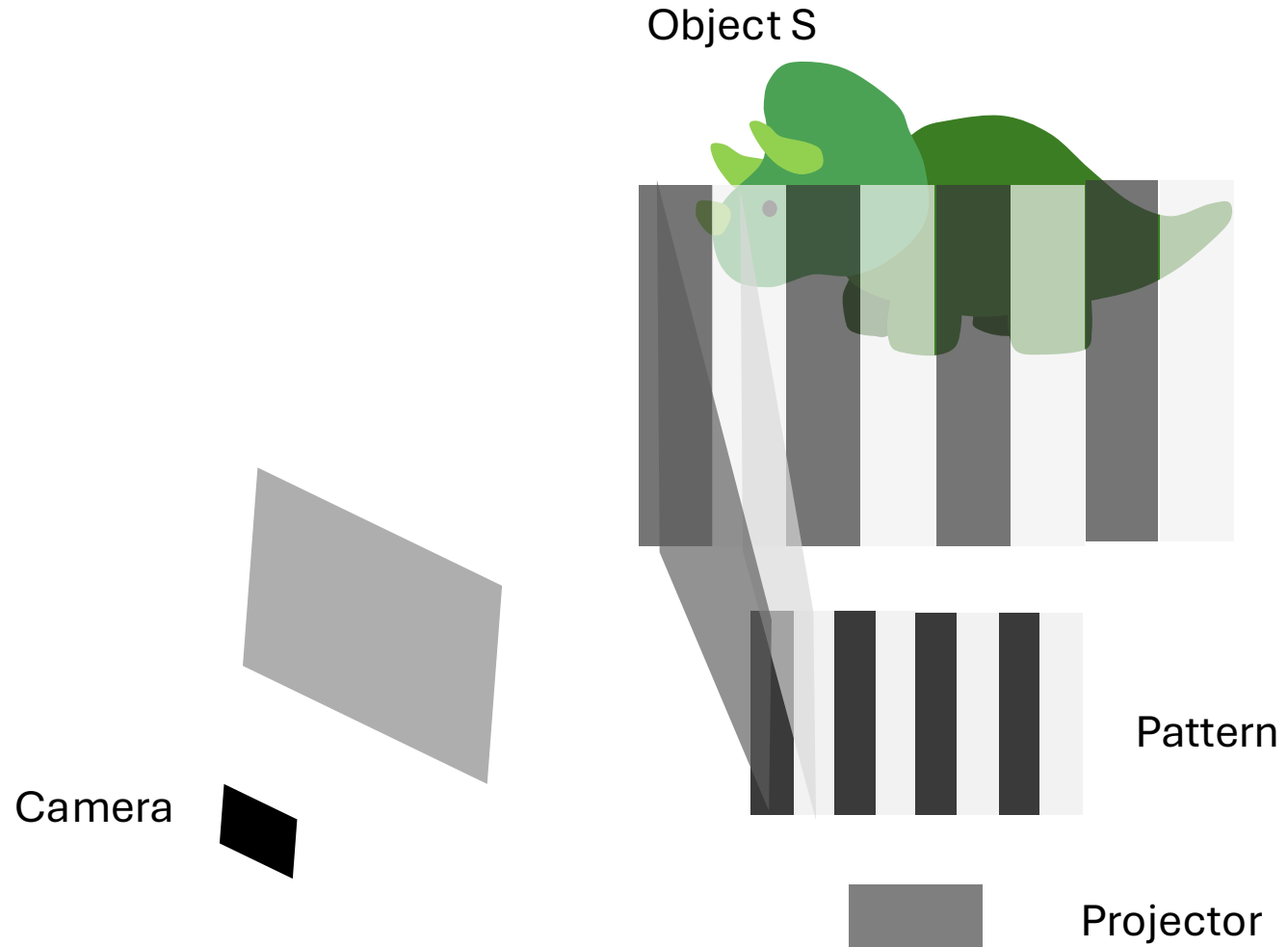
Simple stereo



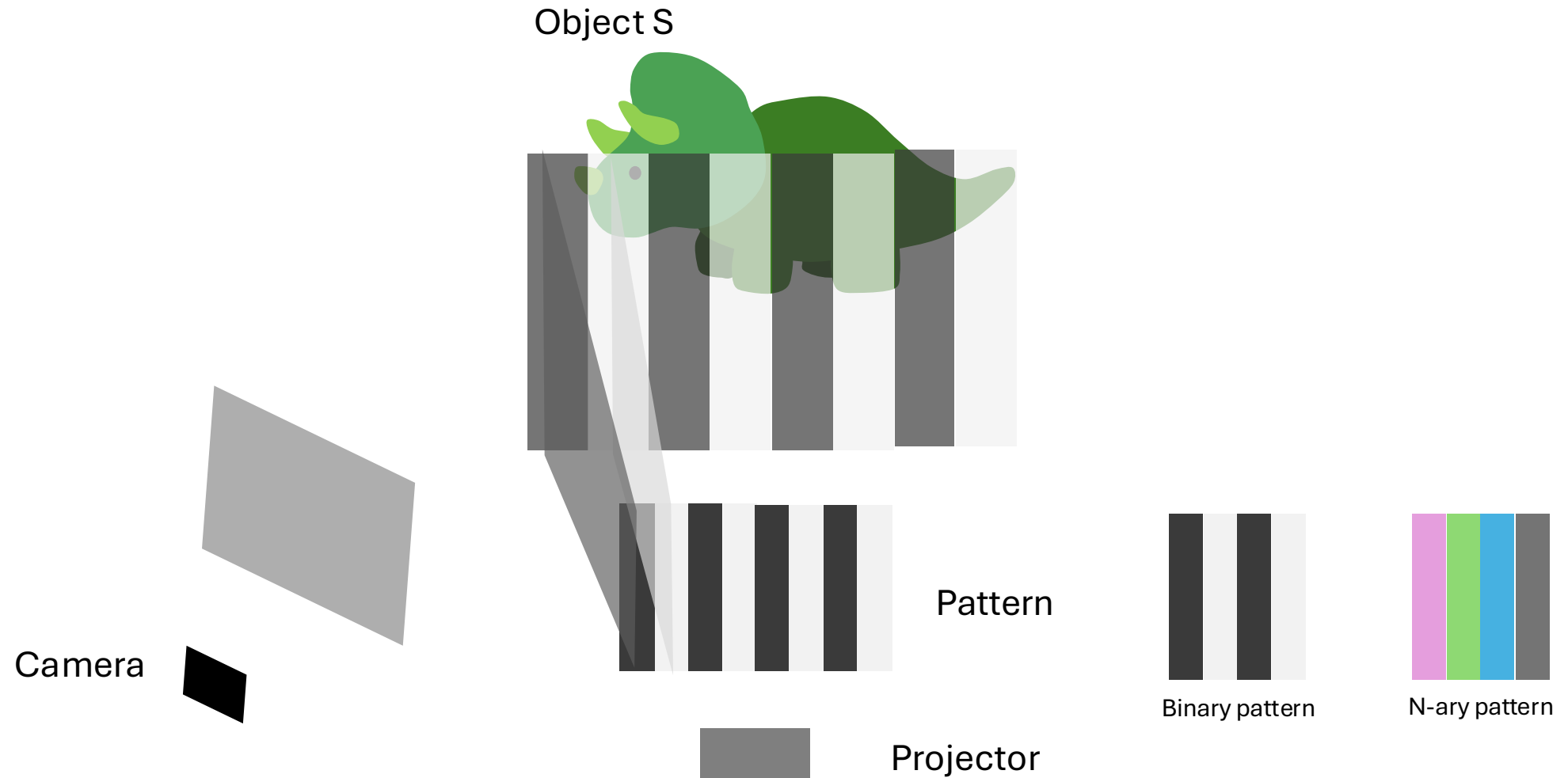
Structured light with one ray of illumination



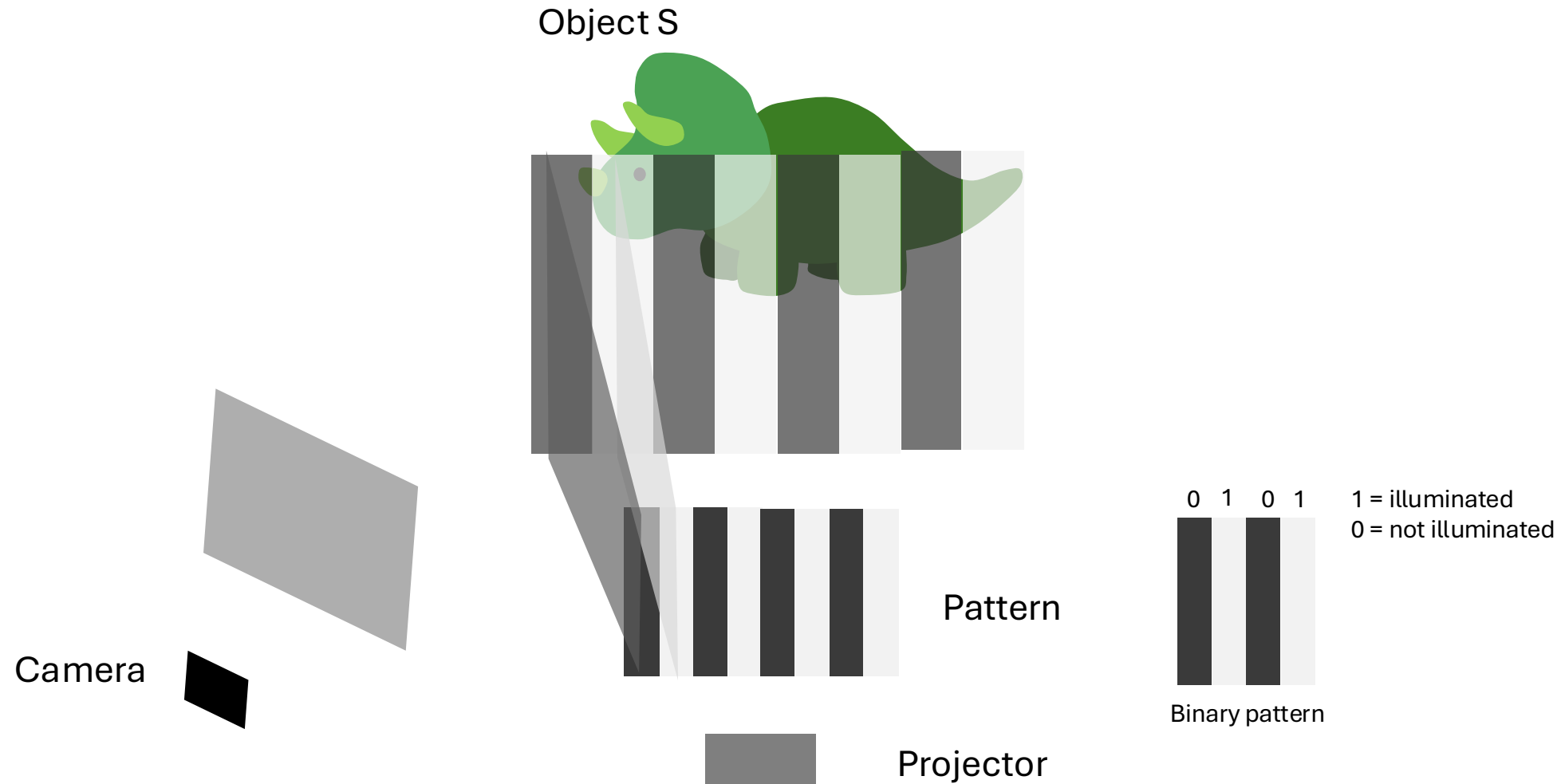
Structured light with stripes of illumination



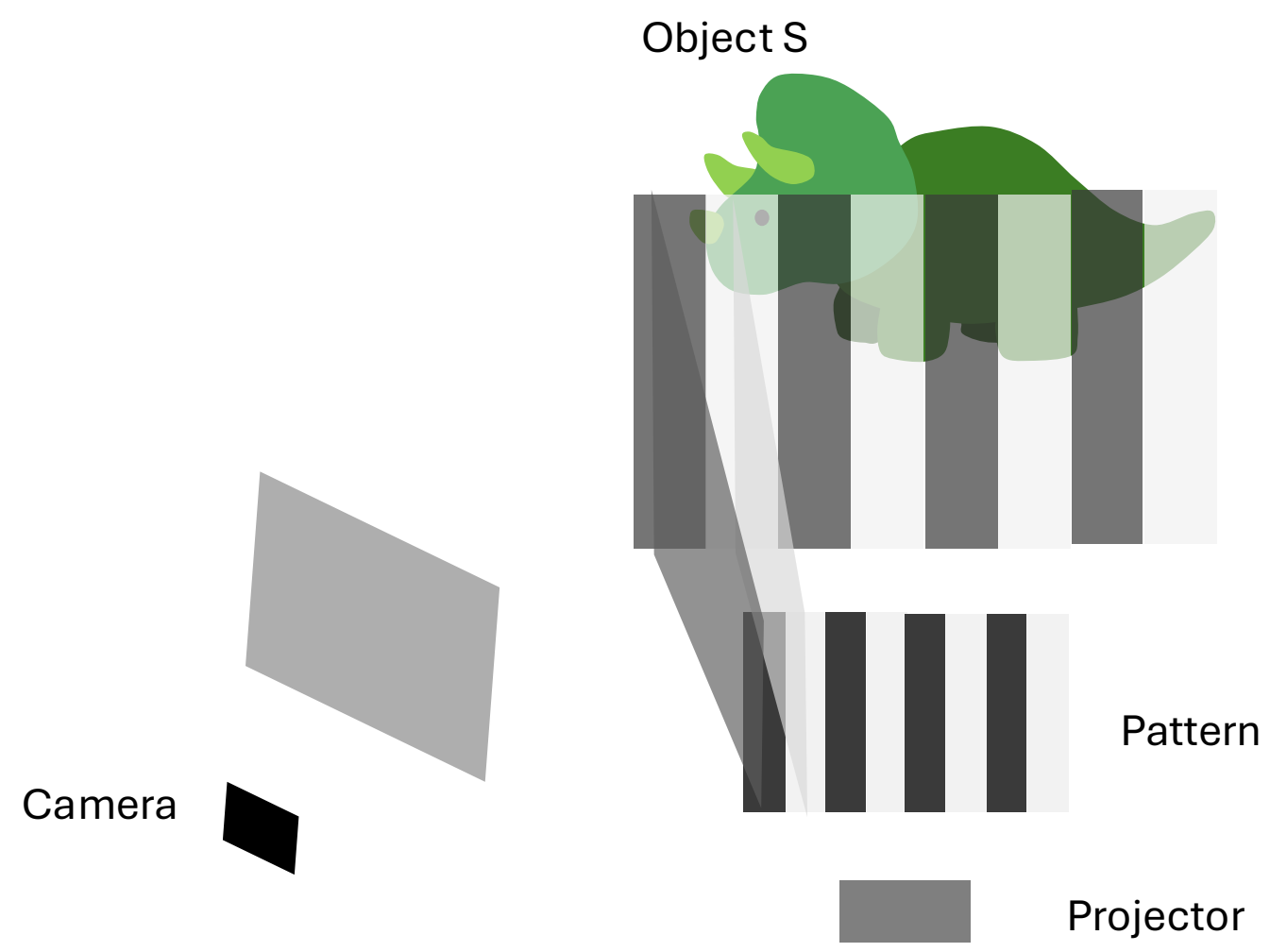
Patterns in structured light



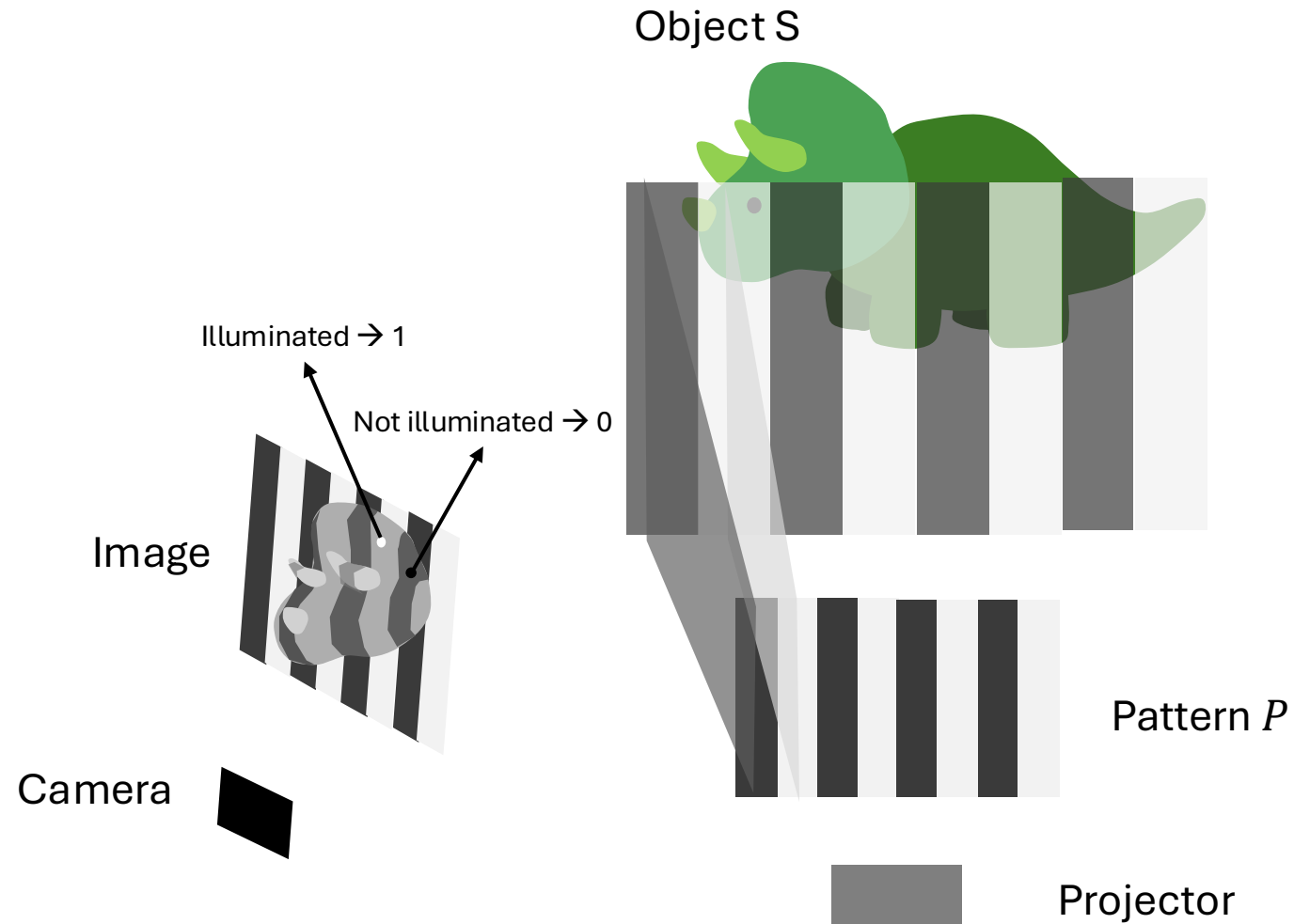
Patterns in structured light



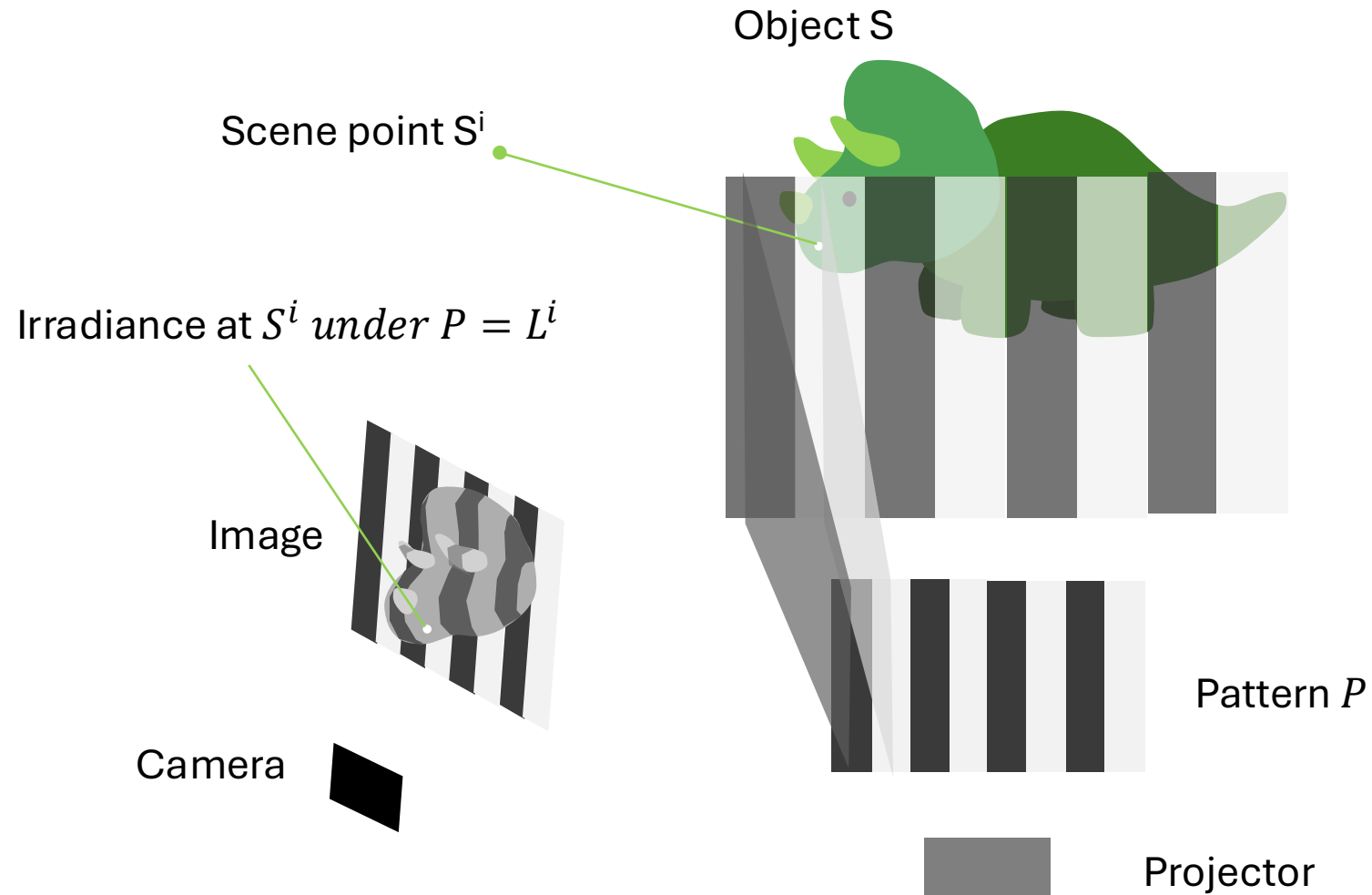
Setup



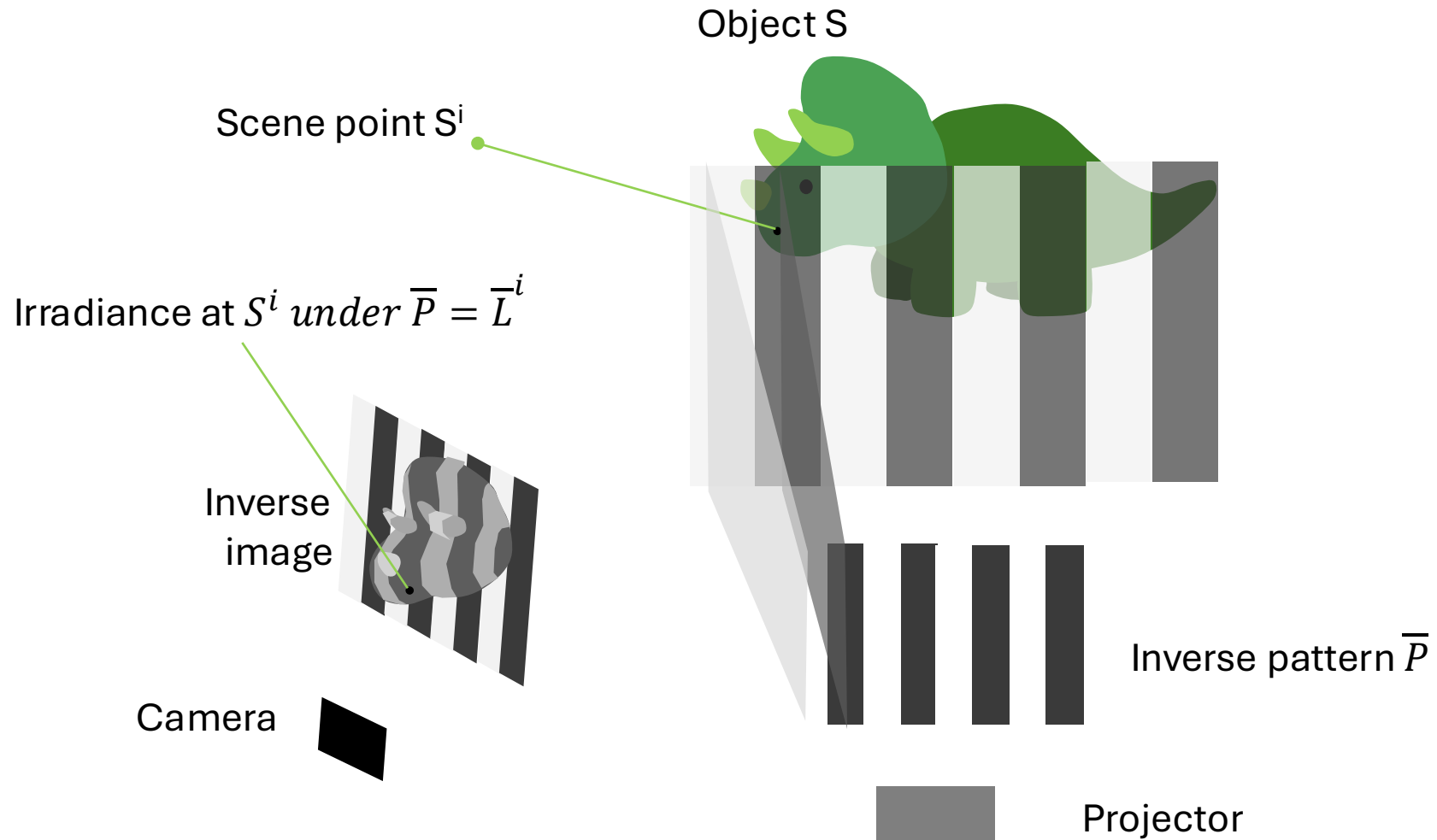
Binarization



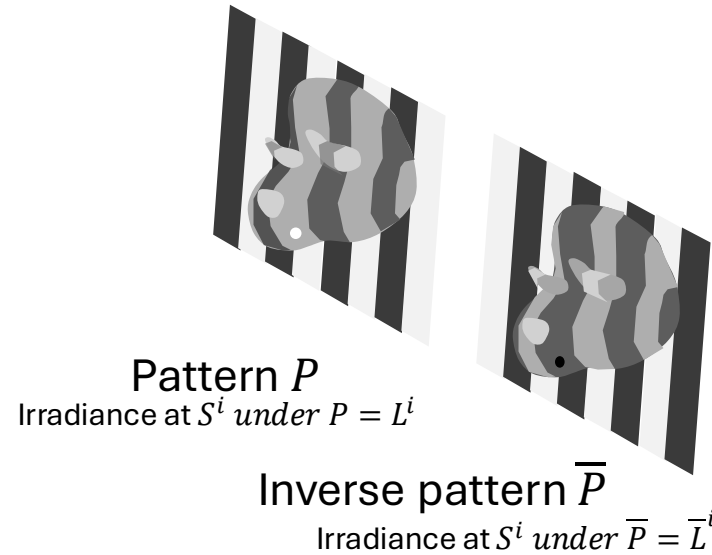
Binarization



Binarization



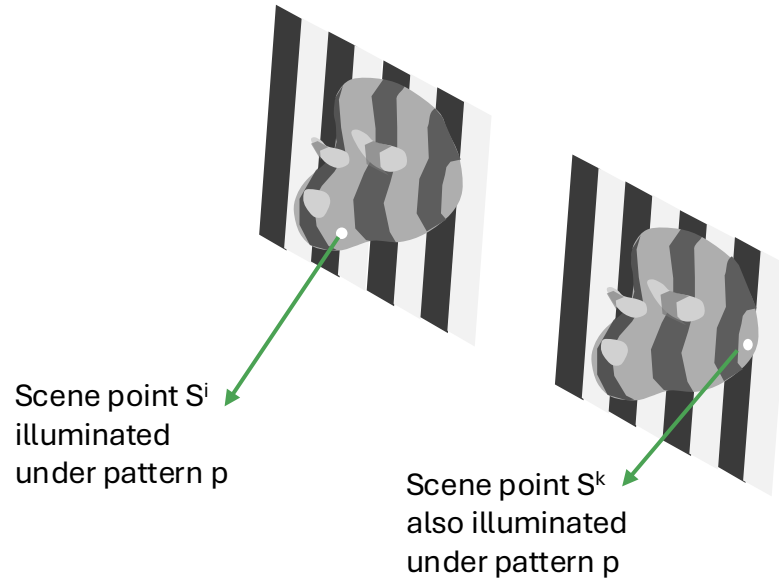
Binarization



To classify a scene point as either 1 or 0:

If $L^i > \bar{L}^i$, then S^i was directly
illuminated $\rightarrow$ classified as 1
($L^i < \bar{L}^i$ classified as 0)

Binarization



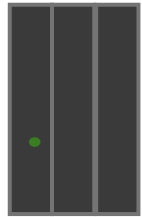
To classify a scene point as either 1 or 0:

If $L^i > \bar{L}^i$, then S^i was directly
illuminated $\rightarrow$ classified as 1
(vice versa for 0)

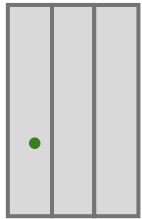
How do we differentiate between the light plane that is illuminating S^i and the other
that is illuminated S^k ?

Project a sequence of patterns

For instance, if • is our scene point and we project three patterns onto it:



Pattern 1



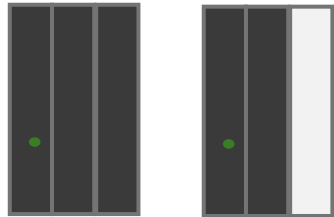
Inverse pattern 1

Since $L^i < \bar{L}^i$, we classify column 1 as 0

Column	1	2	3
Pattern 1			
Code so far	0		

Project a sequence of patterns

For instance, if • is our scene point and we project three patterns onto it:



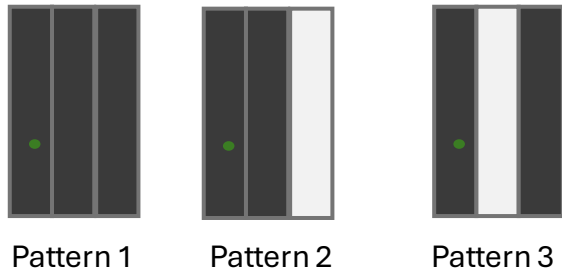
Pattern 1

Pattern 2

Column	1	2	3
Pattern 1			
Pattern 2			
Code so far	00		

Project a sequence of patterns

For instance, if • is our scene point and we project three patterns onto it:

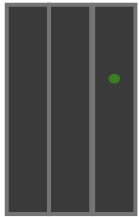


Column	1	2	3
Pattern 1			
Pattern 2			
Pattern 3			
Code so far	000		

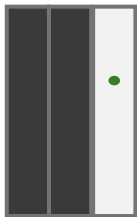
→ This code is going to give a unique label to the column corresponding to S^i

Project a sequence of patterns

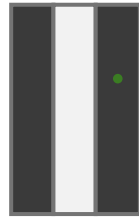
Another example where . is located in a different column



Pattern 1



Pattern 2



Pattern 3

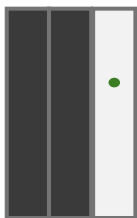
Column	1	2	3
Pattern 1			.
Pattern 2			.
Pattern 3			.
Code so far	010		

Project a sequence of patterns

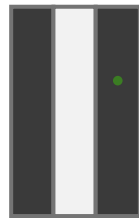
Another example • where is located in a different column



Pattern 1



Pattern 2



Pattern 3

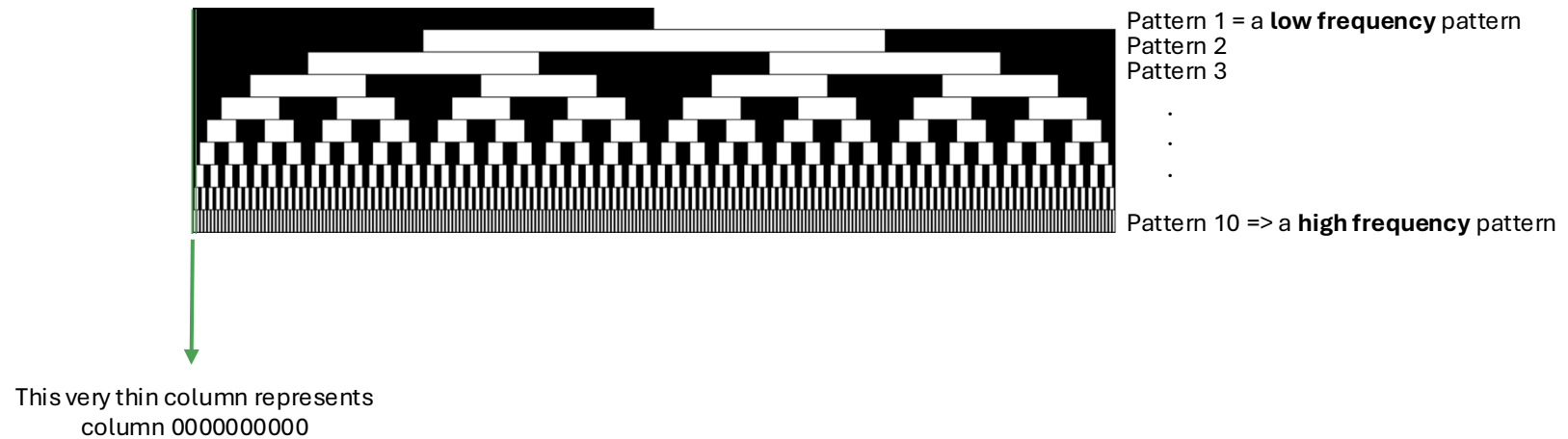
Column	1	2	3
Pattern 1			•
Pattern 2			•
Pattern 3			•
Code so far	010		

Unique code for each column:

Pattern 1			
Pattern 2			
Pattern 3			
Column code	000	001	010

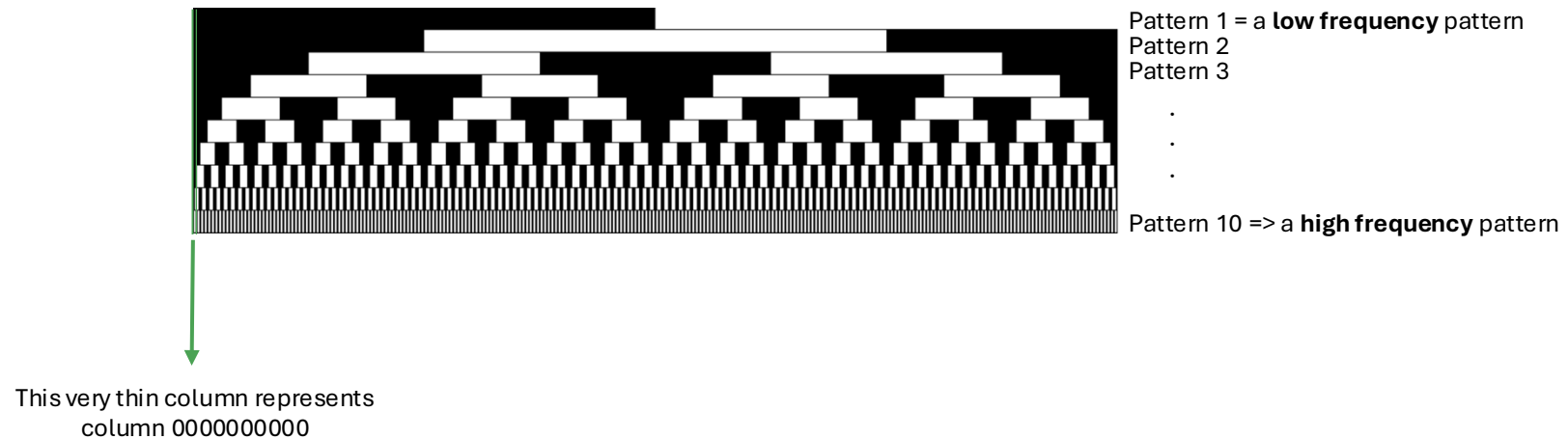
Sequence of patterns: Gray code

Note: 000, 001, 010, 011... is a sequence of codes known as the **gray code**. An extended version of this sequence of patterns that the paper uses is shown below:



Sequence of patterns: Gray code

Note: 000, 001, 010, 011... is a sequence of codes known as the **gray code**. An extended version of this sequence of patterns that the paper uses is shown below:



Now that we can uniquely identify a column, we can know the light plane corresponding to S^i and where it intersects with the camera ray

Circuit Boards

- Now you try to create a 3D model of the circuit board using structured light setup
 - You use conventional gray codes to project 10 patterns onto circuit board

Circuit Boards

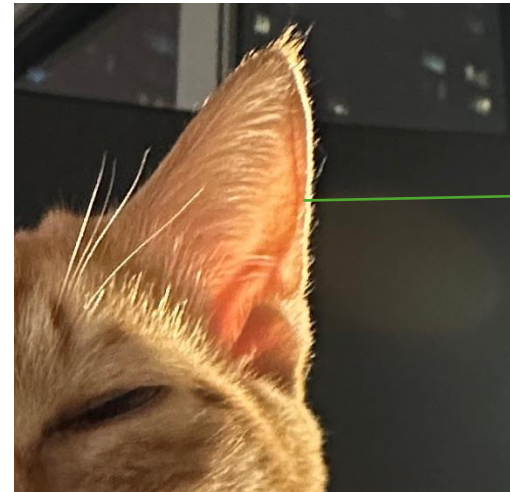
- Now you try to create a 3D model of the circuit board using structured light setup
 - You use conventional gray codes to project 10 patterns onto circuit board
- **Problem!**
 - Your 3D model has high error
 - More specifically, you notice it fails with metallic and translucent materials

Global illumination

- Global illumination = indirect illumination
- Standard structured light assumes each scene point only receives light from direct component
- But in real life, light bounces around!

Types of global illumination

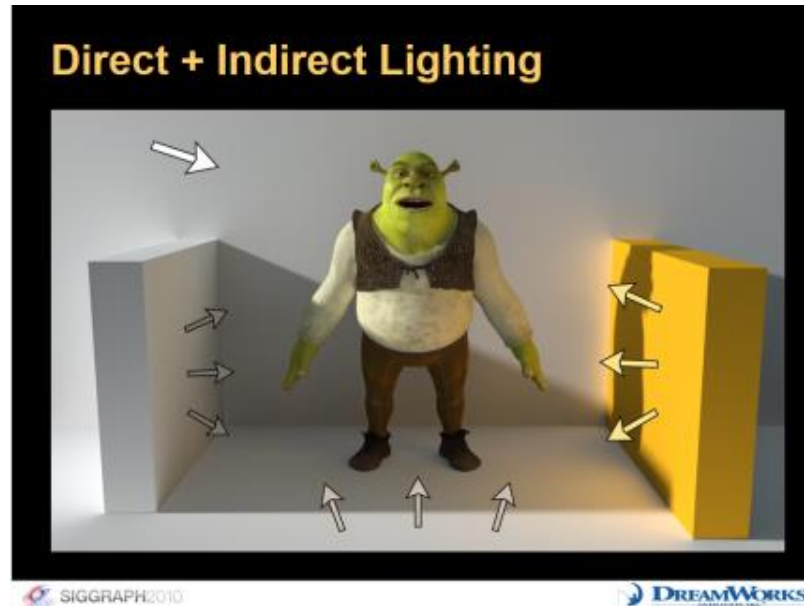
- Short range effects = indirect light that travels only a short distance before reaching a scene point, coming from its immediate neighborhood
 - Seen in subscattering



Light enters the ear,
travels short distance,
and exits through a local
neighbor

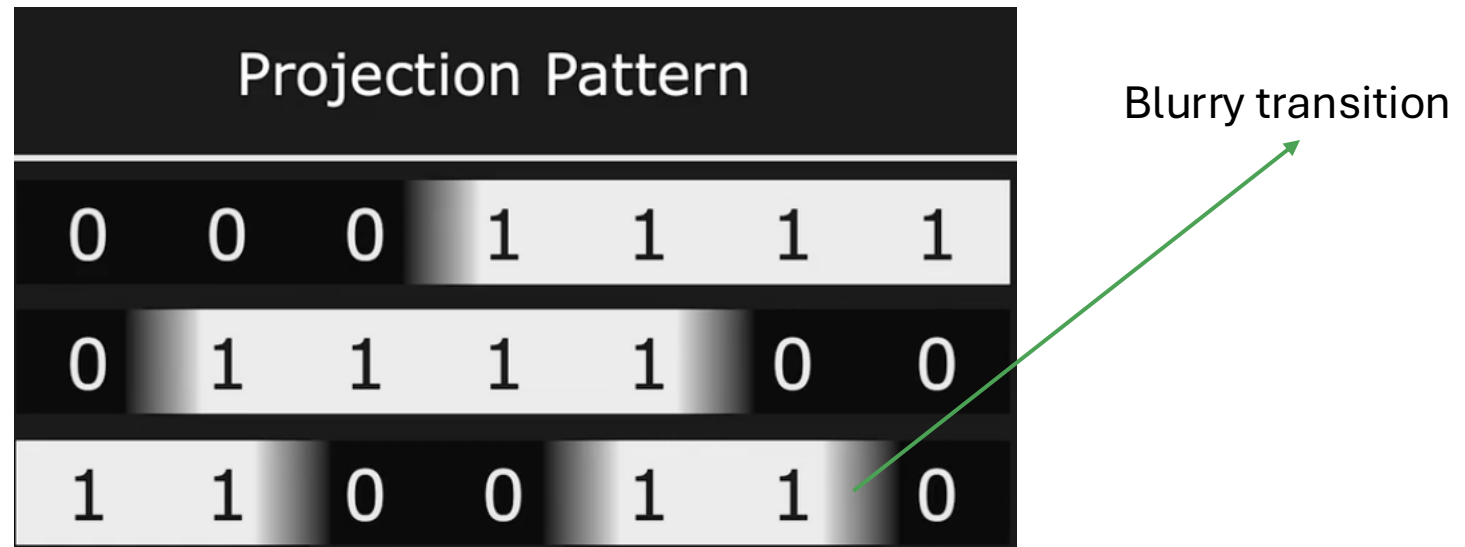
Types of global illumination

- Long range effects = indirect light that travels a large distance before reaching a scene point, coming from further across the scene
 - Seen in inter-reflections and diffusion



Another common problem: Defocus

Projectors don't cast perfectly sharp stripes of light



How global illumination affects projected patterns

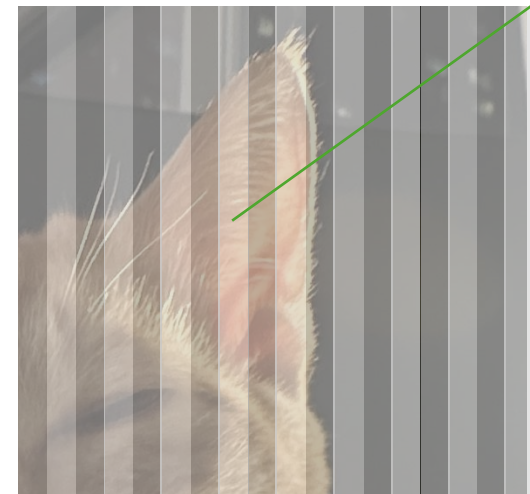
- Short-range effects don't work well with high-frequency patterns (i.e., thin stripes of light)
 - Sub-scattering can weaken irradiance of directly lit scene points, and conversely, strengthen irradiance from global component
 - Blurs the high frequency pattern



Original image



Expected projection of pattern



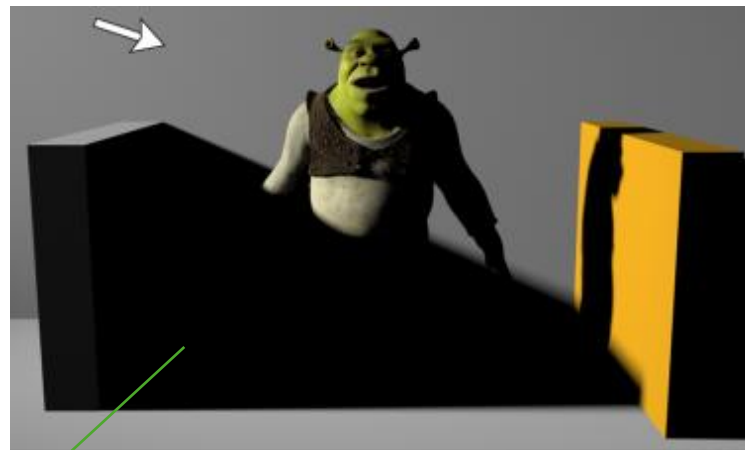
Actual projection of pattern

How global illumination affects projected patterns

- Long-range effects don't work well with low-frequency patterns (i.e., thick stripes of light)
 - Direct light that hits a scene point can bounce onto a scene point that is not directly lit
 - Increases the irradiance of a non-illuminated point, affecting binarization

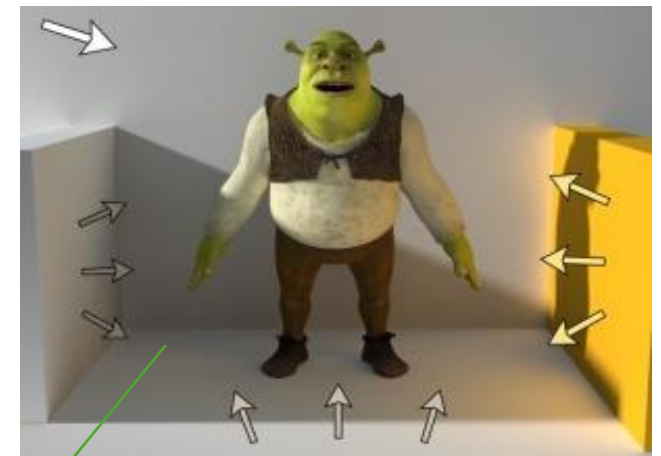


Projected pattern



Expected image

0



Actual image

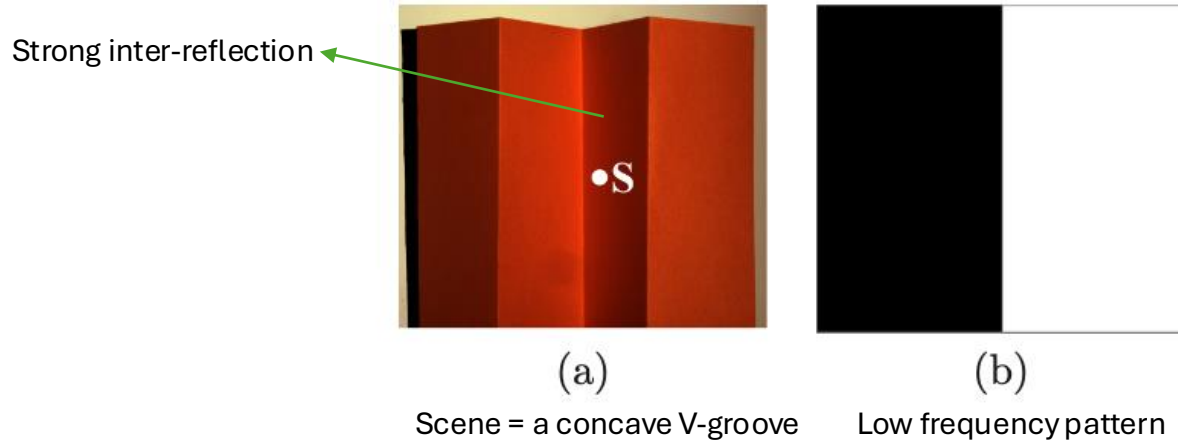
1? 0?

Mathematical explanation

- Originally, structured light assumed:
 - $L^i = L_d^i$ and $L_g^i \approx 0$
 - L_d^i represents the direct component of the irradiance at S^i when the scene is fully lit
 - L_g^i represents the indirect component of the irradiance at S^i when the scene is fully lit
- Now, with global illumination and defocus considered:
 - $L^i = \alpha L_d^i + \beta L_g^i$
 - α represents how much direct light actually reaches S^i
 - β represents the fraction of the global component under pattern P
 - $\bar{L}^i = (1 - \alpha)L_d^i + (1 - \beta)L_g^i$
- For correct binarization (assuming S^i is directly lit):
 - $\alpha L_d^i + \beta L_g^i > (1 - \alpha)L_d^i + (1 - \beta)L_g^i$

Examples from the paper

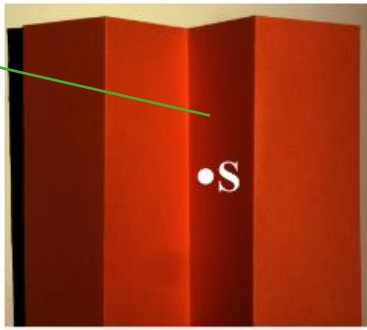
How low frequency pattern affects scene with inter-reflections



Examples from the paper

How low frequency pattern affects scene with inter-reflections

Strong inter-reflection



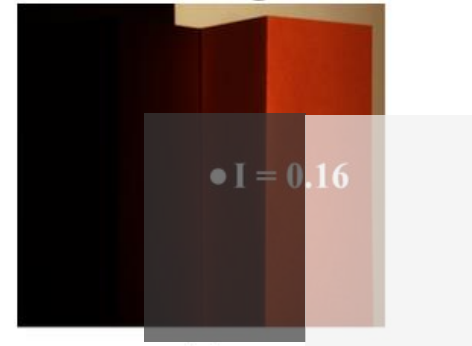
(a)

Scene = a concave V-groove



(b)

Low frequency pattern



(c)

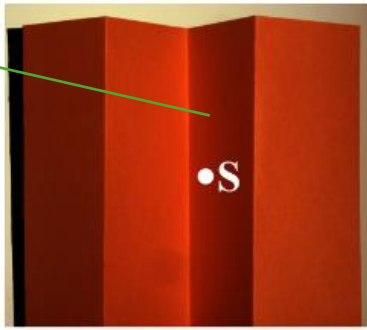
Pattern (b) reflected on (a)

Projector

Examples from the paper

How low frequency pattern affects scene with inter-reflections

Strong inter-reflection



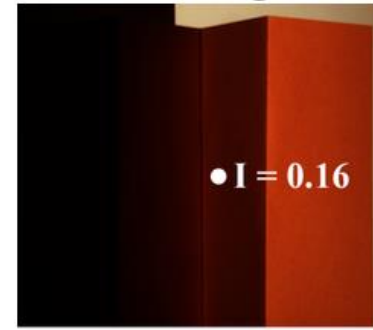
(a)

Scene = a concave V-groove



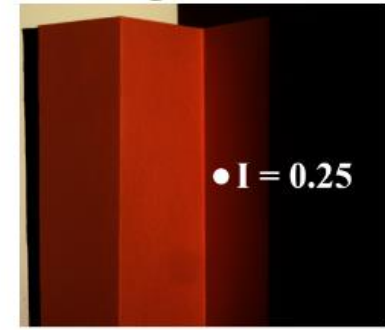
(b)

Low frequency pattern



(c)

Pattern (b) reflected on (a)



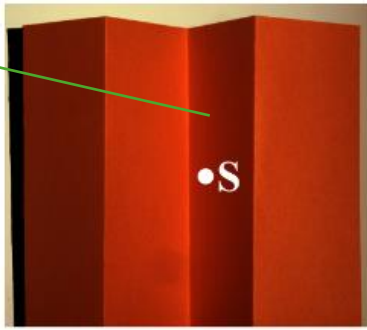
(d)

Inverse pattern $\bar{b}$ reflected on S

Examples from the paper

How low frequency pattern affects scene with inter-reflections

Strong inter-reflection



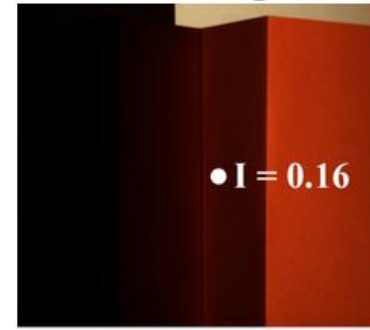
(a)

Scene = a concave V-groove



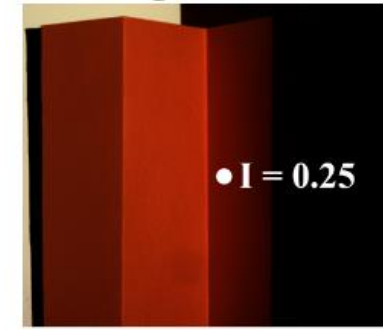
(b)

Low frequency pattern



(c)

Pattern (b) reflected on (a)

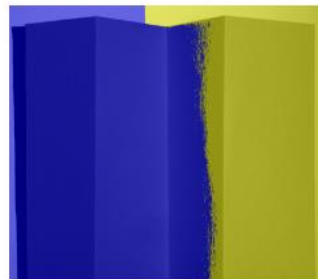


(d)

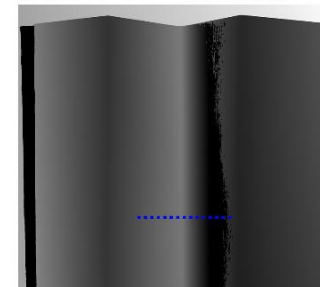
Inverse pattern $\bar{b}$ reflected on S

Since $L^i < \bar{L}^i$, S is classified as 0 (computer assumes that S is not directly lit)

Decoded bit plane
(yellow = directly illuminated;
blue = not illuminated)



(e)



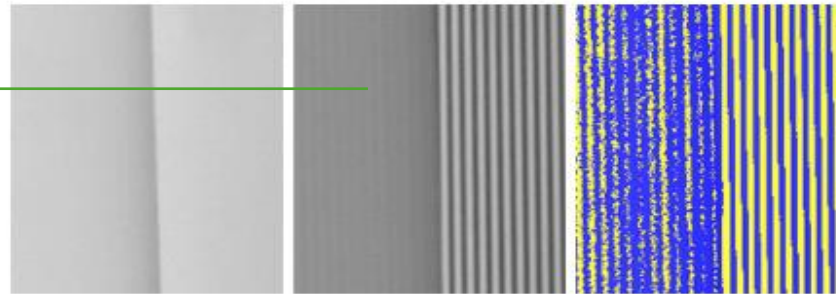
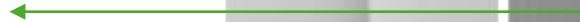
(k) Depth (conventional Gray codes)
Mean absolute error = 28.8mm

Incorrect binarization =>
wrong column code =>
wrong depth

Examples from the paper

How high frequency pattern affects scene with subscattering

Strong subscattering = blurriness



(a)

(b)

(c)

Scene =
On the left: translucent
slab of marble;
On the right = opaque
plane

High frequency
pattern projected
on both materials

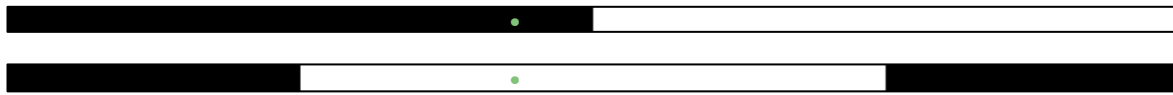
Resulting
binarization

To address errors, authors design a three-stage process:

- Stage 1: Error prevention through carefully designed patterns
 - Most important stage
- Stage 2: Error detection
- Stage 3: Error correction

Pattern designs for long-range effects

- Conventional gray code:
 - Low frequency patterns are projected first to make broad distinctions between columns (determines classification of higher order bits)



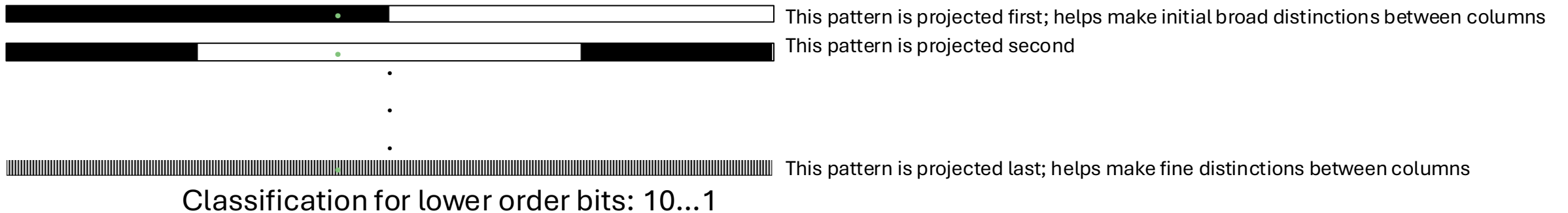
This pattern is projected first; helps make initial broad distinctions between columns

This pattern is projected second

Classification for higher order bits: 10

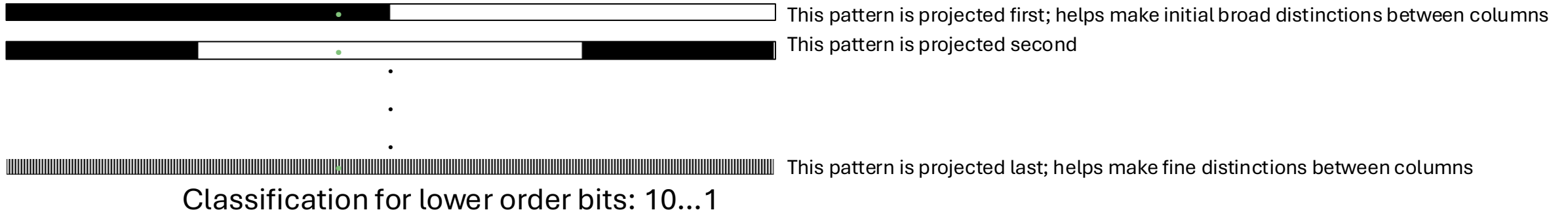
Pattern designs for long-range effects

- Conventional gray code:
 - Low frequency patterns are projected first to make broad distinctions between columns
 - High frequency patterns are projected later in the sequence to make finer distinctions



Pattern designs for long-range effects

- Conventional gray code:
 - Low frequency patterns are projected first to make broad distinctions between columns
 - High frequency patterns are projected later in the sequence to make finer distinctions
- If misclassifications are made in the earlier parts of the sequence (i.e., wrong binarization when projecting low frequency patterns), this can cause incorrect binarization of higher bits

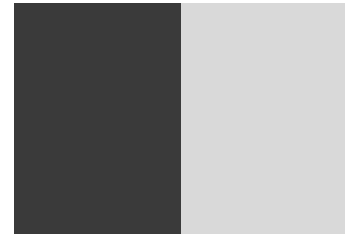


How to make initial broad distinctions with scenes that have strong inter-reflections?

- Important observation authors make: the ultimate goal is to get the correct binarization of the captured image
- Can we make correct binarizations of higher order bits without actually projecting low frequency patterns?
- Is there an operator P_{hf}^2 ? $P_{hf}^1 = P_{1f} \Rightarrow P_{hf}^2 = P_{1f} ? P_{hf}^1$
 - P_{1f} is a low frequency pattern
 - P_{hf}^1 and P_{hf}^2 are the high frequency patterns

To help you visualize:

- Suppose we have our low frequency pattern

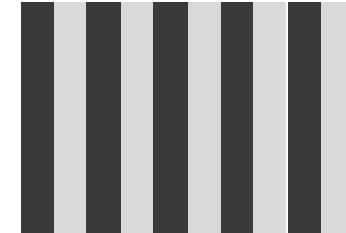


To help you visualize:

- Suppose we have our low frequency pattern

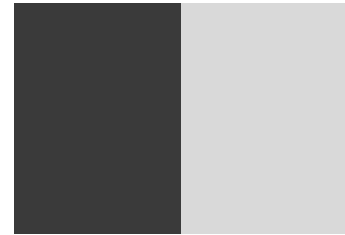


- and we have some high frequency base pattern

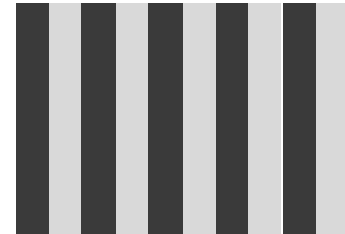


To help you visualize:

- Suppose we have our low frequency pattern



- and we have some high frequency base pattern



- Can we decompose our low frequency pattern into two high frequency patterns (one of which is the base pattern) and can we then compose that resulting pattern with our base pattern to get back our low frequency pattern?

XOR operator

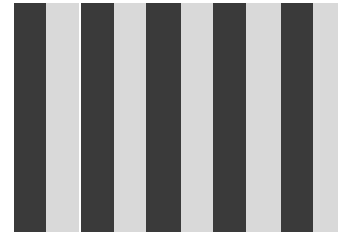
- $P_{hf}^2 \oplus P_{hf}^1 = P_{1f} \Rightarrow P_{hf}^2 = P_{1f} \oplus P_{hf}^1$
- Then, we can compute the binarizations under the two high frequency patterns and find the correct binarization under the low frequency pattern

XOR operator example

When coding:



$\oplus$



$$0 \oplus 0 = 0$$



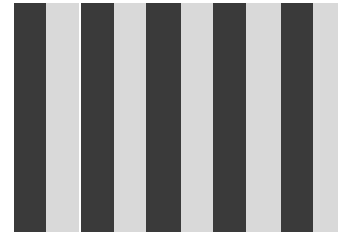
0

XOR operator example

When coding:



$\oplus$



$$0 \oplus 1 = 1$$



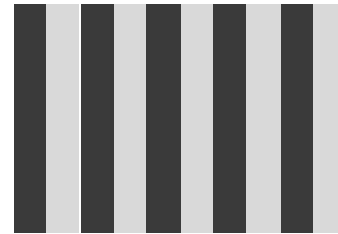
0 1

XOR operator example

When coding:



$\oplus$



$$1 \oplus 1 = 0$$



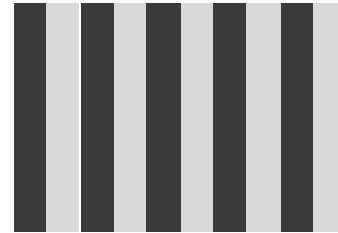
0 1 0 1 0 0 1 0 1 0

XOR operator example

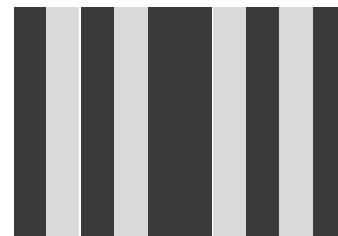
When coding:



$\oplus$



=



0 1 0 1 0 0 1 0 1 0

→ Project this onto scene instead of our low frequency pattern

Note: no overhead involved
in this approach

XOR operator example

When decoding:



Use this to make correct binarization of higher order bits

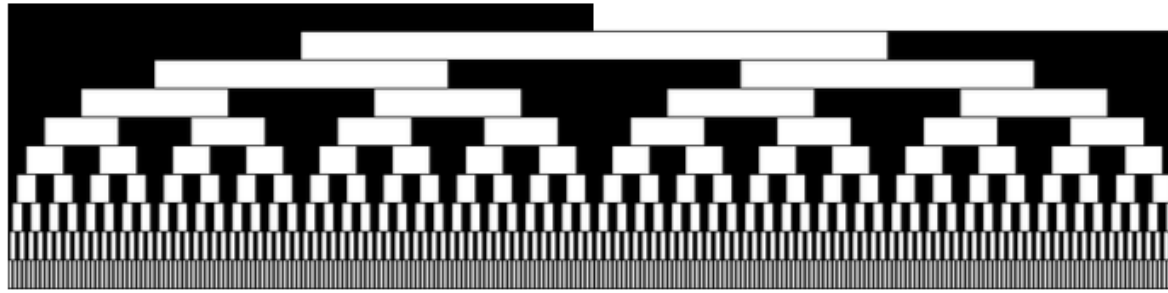
=



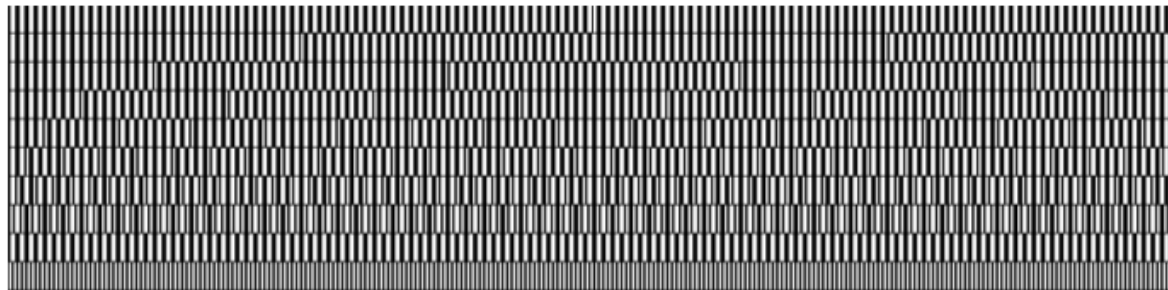
$\oplus$



XOR pattern example from paper



Conventional Gray Codes



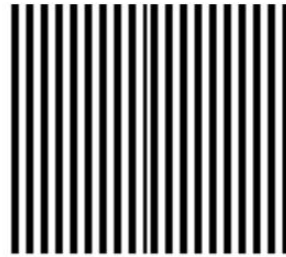
XOR-04 Codes

XOR operator on long-range effects



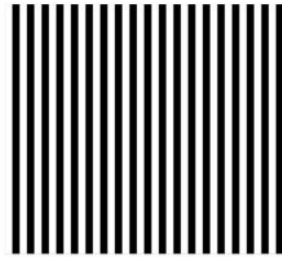
(b)

Original low
frequency
pattern



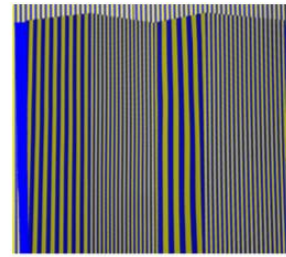
(f)

First high
frequency
pattern for (b)



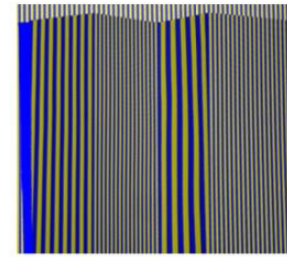
(g)

Second high
frequency
pattern for (b)



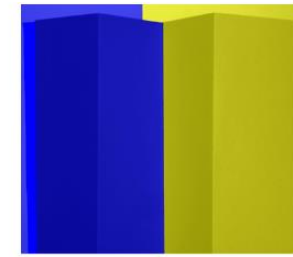
(h)

Binarization of
image with
pattern (f)



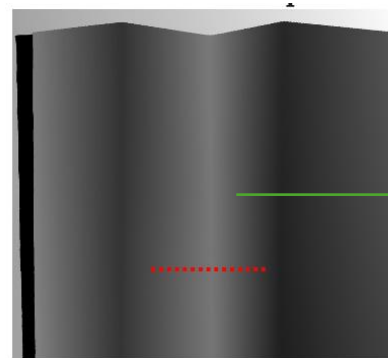
(i)

Binarization of
image with
pattern (g)



(j)

Binary decoding



No more wrong depth value from inter-reflections

(l) Depth (our XOR04 codes)
Mean absolute error = 1.4mm

Patterns for short-range effects

- Scenes with subscattering can work well with low frequency patterns
- Conventional gray codes may work with short-range effects initially...
 - But as we project the later patterns in the sequence (i.e., high frequency patterns), we can't make finer distinctions between columns (incorrect binarization of lower-order bits)

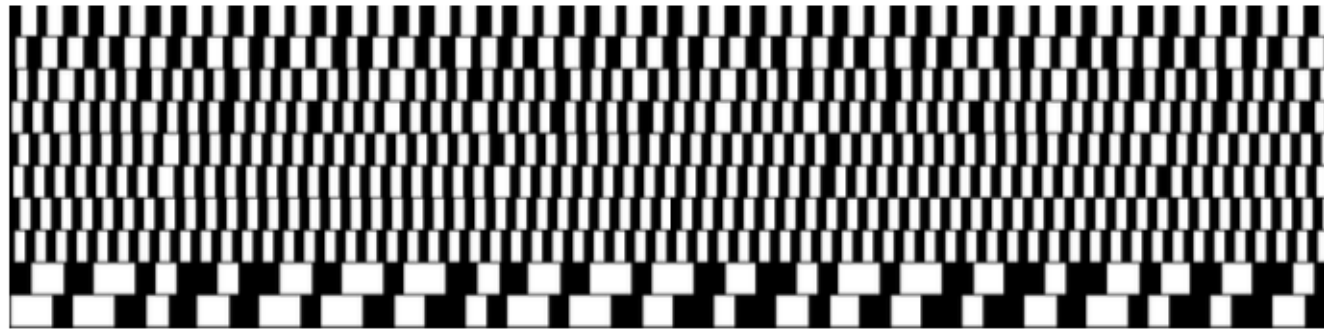
Patterns for short-range effects

- Scenes with subscattering can work well with low frequency patterns
- Conventional gray codes may work with short-range effects initially
 - But as we project the later patterns in the sequence (i.e., high frequency patterns), we can't make finer distinctions between columns (incorrect binarization of lower-order bits)
- Authors ask: Is there a large minimum stripe-width?
 - That is, how large can our thinnest stripe-width be and still help make correct binarizations of lower-order bits?

Patterns for short-range effects

- Brute force approach = impossible
- Goddyn et al. uses max min-SW of 8 works well
 - For reference, conventional gray codes have max min-SW of 2

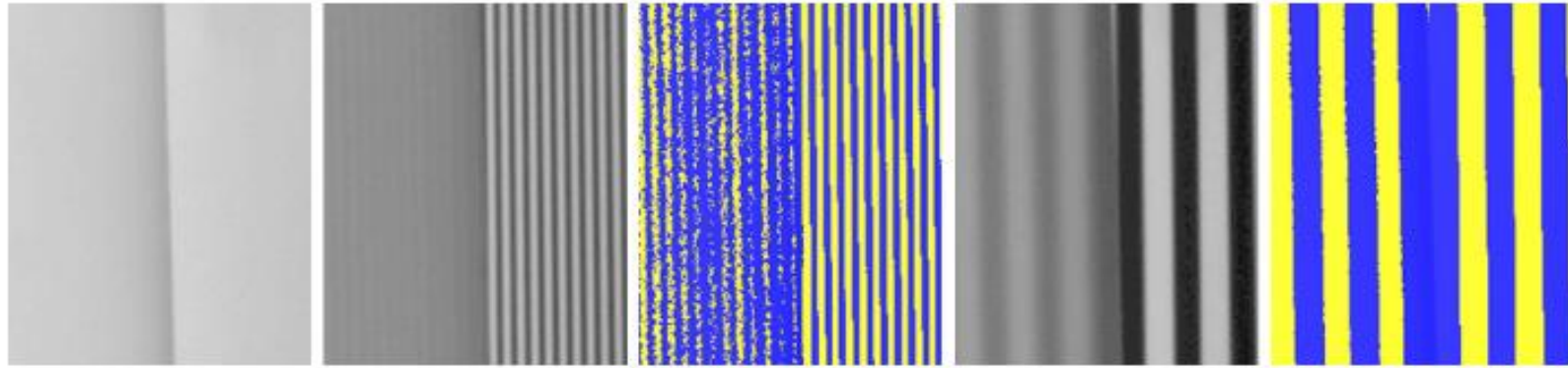
The 8 max min-SW used by paper



Gray codes with maximum min-SW

Stripe width range: [8,32]

Patterns for short-range effects



(a)

Scene

(b)

High
frequency
pattern on
scene

(c)

Binarization
under high
frequency

(d)

Low
frequency
pattern on
scene

(e)

Binarization
under low
frequency

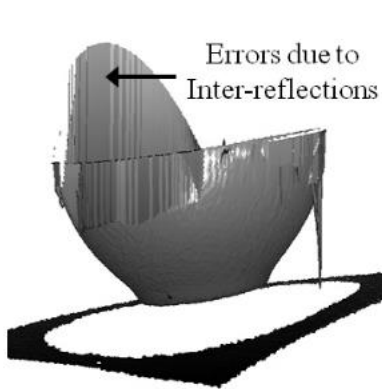
Patterns for both long and high-range effects

- Ensemble of patterns
 - Project 4 sets of patterns:
 - For long-range
 - 1 set of XOR-04
 - 1 set of XOR-02
 - For short-range
 - 1 set of conventional gray
 - 1 set of max min-SW
- If any two agree on depth value within a small threshold, that value is returned
- If only two gray codes agree (i.e., patterns for short-range), then use value from max min-SW (better depth resolution)

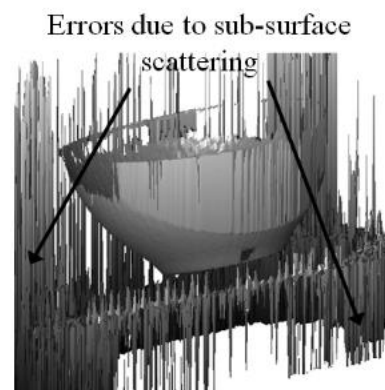
Ensemble results



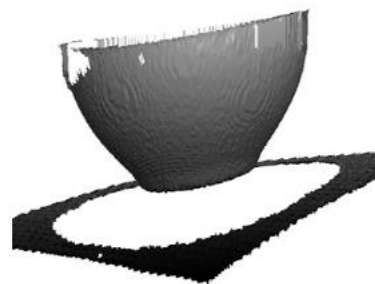
(a) Bowl on a translucent marble slab



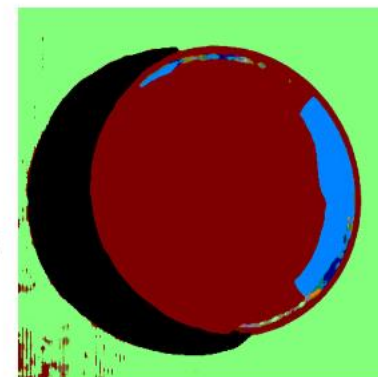
(b) Conventional Gray codes (11 images)



(c) Modulated phase shifting [4] (162 images)



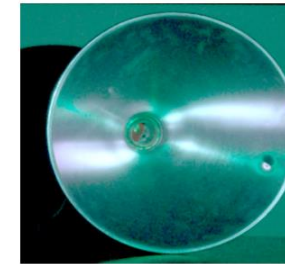
(d) Our ensemble codes (41 images)



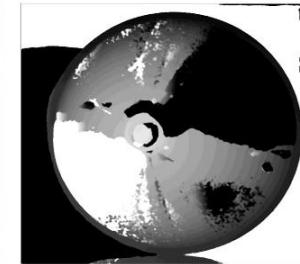
(e) Error map for codes

Error detection & correction

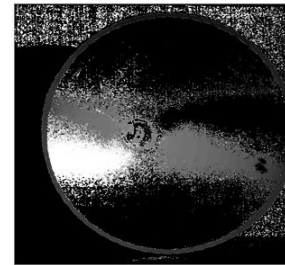
- Error detection: If none of the four patterns agree, classify that pixel as an error pixel
- Error correction:
 1. Find scene points of the corresponding error pixels
 2. Only illuminate those scene points
 3. Do 2 iterations of XOR-02 and XOR-04Idea: less global light corrupting the points



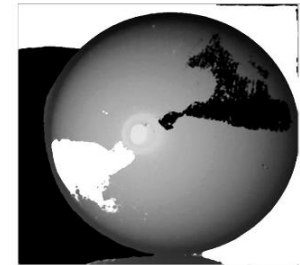
(a) Metallic lamp



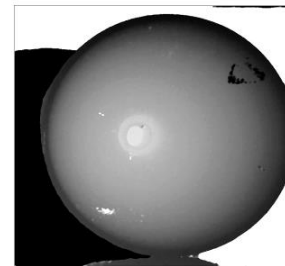
(b) Conventional Gray (11 images)



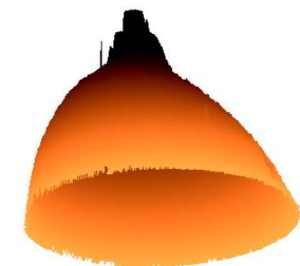
(c) Modulated phase shifting [4] (162 images)



(d) Our ensemble codes (41 images)



(e) After error detection and correction (81 images)



(f) 3D visualization of (e)

Thank you!

